

Notes: Delabastita and Rubens (JPE, 2025)

Colluding against Workers

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1 Big picture

- **Question.** How can we detect and quantify **employer collusion** in labor markets when wage markdowns may also come from ordinary oligopsony, search frictions, employer differentiation, or other noncollusive sources?
- **Setting.** The paper develops a method that combines a **production-cost model** with a **labor supply model**, then applies it to **227 Belgian coal firms** in Liège and Namur between **1845 and 1913**.
- **Core challenge.** A large wedge between the marginal revenue product of labor and wages does *not* by itself prove collusion. It could come from noncooperative oligopsony power. So the paper needs a way to separate the **collusive** part of wage markdowns from the **noncollusive** part.
- **Main methodological contribution.** The paper:
 1. estimates wage markdowns from production and cost data **without imposing labor-market conduct assumptions**;
 2. derives the markdown that would arise under **no collusion** and under **full collusion** from a labor supply model;
 3. compares the actual markdown to those two bounds to infer the degree of collusion.
- **Why it matters.** Policy depends on the source of labor market power. If wage suppression comes from search frictions or employer differentiation, the policy response differs from a world where firms are **colluding**. Antitrust is central in the second case.
- **Why this setting is useful.** Nineteenth-century Belgium is attractive because cartels were legal and therefore historically observable; the coal industry has a long firm-level panel; and coal is relatively homogeneous, which simplifies the empirical design.
- **Main empirical question.** How much of the rise in Belgian coal firms' wage markdowns was due to **collusion**, especially around the **1897 Liège coal cartel**?
- **Main empirical results.**
 - Up to the 1870s, wage markdowns were fairly stable: at the median firm, workers received about **two-thirds** of their marginal product.
 - In the 1880s–1890s, markdowns rose further, leaving workers with around **60%** of marginal product.
 - After 1900, markdowns rose sharply, leaving workers with **less than 50%** of marginal product.
 - The pre-1900 increase was mostly due to **noncollusive** forces.
 - The post-1900 jump was driven **entirely by collusion** and lines up with the 1897 cartel.
 - Relative to observed precartel conduct, the cartel lowered **wages and employment by 6%–17%**.
 - Relative to a world with no labor-market collusion at all, the cartel lowered **wages and employment by 10%–25%**.
- **Economic takeaway.** A downstream cartel can create substantial **upstream labor-market power**. Even when product markets are fairly competitive, collusion on output can sharply increase wage markdowns.

2 Data and measurement (key objects)

2.1 Data sources and sample

- **Main source.** Annual inspection reports of the *Administration des Mines*, a Belgian state agency that inspected all coal mines.
- **Coverage.** The paper uses the exceptionally well-preserved archives for the provinces of **Liège** and **Namur**.
- **Main unit.** The data are at the **mining concession** level. The paper argues that, in practice, this is a good proxy for the firm level because multiple-concession ownership was rare in these provinces and often followed by merger into a single concession.
- **Observed variables.**
 - annual coal extraction in tons, by coal type,
 - mine-gate coal prices,
 - employment in workers and days,
 - underground versus surface labor,
 - *ordinary expenses* = intermediate input expenditure,
 - *extraordinary expenses* = fixed capital investment,
 - horsepower of machines up to 1899.
- **Capital.** The capital stock is constructed using a **perpetual inventory method**.
- **Additional sources.**
 - employers' association membership from digitized monthly bulletins,
 - cartel membership from De Leener's cartel lists,
 - railroad and tramway connections,
 - CPI-based deflators.

2.2 Markets, firms, products, and market definitions

- **Firms.** 227 coal firms appear in the sample.
- **Geographic scope.** Liège and Namur together accounted for roughly **30% of Belgian coal workers** and **20%–25% of coal output**.
- **Average number of firms per year.** About **60** in Liège and **19** in Namur.
- **Product market.** Coal is treated as a **homogeneous product** in the baseline model. Output is summed across coal quality categories because caloric-content differences are argued not to affect mining productivity.
- **Labor market.** In estimation, labor markets are defined at the **municipality-year** level.
- **Key support for that labor-market definition.**

- Around **90%** of workers commuted less than **10 km**.
- Municipality-year fixed effects explain about **93%** of miner wage variation; adding firm fixed effects raises this only to **94%**.

This supports the modeling choice that wages are essentially the same within a local labor market in a given year.

2.3 Coal production and technological environment

- **Production steps.**

1. reach the coal vein by digging a shaft;
2. extract coal manually with pickaxes;
3. haul coal to the surface;
4. sort coal from debris.

- **Mechanization.**

- Haulage was already partly mechanized early in the sample.
- Water pumps and ventilation fans diffused mainly from the 1870s onward.
- **Coal cutting machines**, the main labor-saving mining technology elsewhere, were barely adopted in this setting because the coal veins were too narrow.

- **Why this matters for the model.** The weak diffusion of cutting machines is one reason the authors view **factor-biased technical change** as less likely to explain the big post-1897 change in markdowns.

2.4 Labor relations and collusion institutions

- **Labor relations.**

- Belgium was a low-wage, laissez-faire economy.
- Worker coalitions were restricted for much of the nineteenth century.
- Trade unions remained weak until much later.
- Hiring and firing costs were minimal.
- Coal workers were mobile across employers.

- **Employers' associations.**

- These associations coordinated industry decisions and explicitly helped “**coordinate salary fluctuations.**”
- About **33%** of firms in the data belonged to an employers' association, but those firms produced about **80%** of output.
- They seem to have been important informal vehicles for labor-market collusion.

- **Coal cartel.**

- In **1897**, 27 Liège coal firms formed the *Syndicat de Charbonnages Liégeois*.
- The cartel imposed output quotas and fines for violating them.
- It controlled about **75%–80%** of Liège-basin sales.
- Although the cartel did not directly set employment quotas, reducing output mechanically reduced labor demand.

2.5 Key descriptive facts

- **Output per worker and wages.**

- From 1845 to about 1875, wages and output per worker moved together.
- In the late 1870s and 1880s, wages flattened while output per worker rose.
- In the late 1890s, wages rose while output per worker fell.

- **Labor cost share.**

- Median labor cost share was roughly stable until the 1890s.
- After 1900, both median and weighted-average labor cost shares fell.

- **Coal prices around the cartel.**

- Before 1897, Liège coal prices were below the import price of coal.
- After the cartel started, Liège coal prices rose toward the import-price ceiling.

Interpretation. Raw descriptive patterns already suggest a rising wedge between labor productivity and wages around 1900, but the paper argues that these raw series are not enough: one needs a production model to separate labor's marginal product from changes in capital, materials, and product-market conditions.

2.6 Unobservables and timing

- **Productivity.** Firm-level productivity is unobserved and follows an AR(1) process.

- **Labor supply shifter.** Local labor markets have an unobserved attractiveness component.

- **Timing.**

- At $t - 1$, firms choose capital investment and decide whether to collude.
- At t , after observing the productivity shock, firms choose labor and materials.

Economic meaning. Labor and materials are treated as **static variable inputs**; capital is treated as a **dynamic fixed input**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production function

The baseline production function is Cobb–Douglas in labor, materials, and capital:

$$q_{ft} = \beta_l l_{ft} + \beta_m m_{ft} + \beta_k k_{ft} + \omega_{ft}.$$

- q_{ft} : log coal output
- l_{ft} : log effective labor
- m_{ft} : log intermediate inputs
- k_{ft} : log capital stock
- ω_{ft} : log TFP

Interpretation. Output depends on labor, materials, and capital in a standard way, and the productivity term captures firm-year efficiency not explained by observed inputs. The authors prefer Cobb–Douglas as a tractable baseline and later check robustness with more flexible specifications.

3.2 Productivity process and timing

Productivity evolves as

$$\omega_{ft} = \rho \omega_{f,t-1} + u_{ft}.$$

- u_{ft} is the unexpected productivity shock.
- ρ measures persistence.

Interpretation. This AR(1) structure is not only an economic assumption about productivity persistence; it is also part of the identification strategy. It lets the paper estimate production parameters without inverting input demand under latent markups and markdowns.

3.3 Labor supply

At the market level, labor supply is

$$W_{it}^l = L_{it}^{\Psi^l} \nu_{it}.$$

- W_{it}^l : market wage
- L_{it} : market employment
- Ψ^l : inverse labor supply elasticity

- ν_{it} : local labor-supply shifter

Interpretation.

- If $\Psi^l = 0$, labor supply is perfectly elastic and firms are wage takers.
- If $\Psi^l > 0$, labor supply is upward sloping and employers have labor-market power.
- The paper assumes homogeneous employers within a local market, consistent with the near-absence of within-town wage dispersion.

3.4 Markdowns and markups

The **wage markdown** is

$$m_{ft}^l = \frac{\text{MRPL}_{ft}}{W_{ft}^l},$$

and the percent markdown below marginal revenue product is

$$d_{ft}^l = \frac{m_{ft}^l - 1}{m_{ft}^l}.$$

The **price markup** is the coal price over marginal cost,

$$m_{ft} = \frac{P_{ft}}{MC_{ft}}.$$

Interpretation.

- $m_{ft}^l > 1$ means workers are paid below their marginal revenue product.
- $m_{ft} > 1$ means the firm charges price above marginal cost.

The paper's key point is that labor-market power and product-market power are separate objects and need not move together.

3.5 Employer conduct: no collusion, full collusion, and the conduct parameter

The firm chooses inputs to minimize a weighted combination of its own and rivals' costs:

$$\min_{L_{ft}, M_{ft}} \sum_{g \in F_i} \lambda_{fgt} (L_{gt} W_{gt}^l + M_{gt} W_{gt}^m) - MC_{ft} (Q(L, M, K, \Omega; \beta) - Q_{ft}).$$

- $\lambda_{fgt} = 1$ for own cost.
- For rivals, $\lambda_{fgt} \in [0, 1]$ captures the extent to which firm f internalizes firm g 's costs.

Interpretation. This nests:

- **no collusion:** firms minimize only their own costs;
- **full collusion:** firms minimize joint costs as if they were one firm.

3.5.1 No collusion

Under noncooperative labor demand, the markdown is

$$m_{ft}^{\text{NC}} = 1 + s_{ft}^l \Psi^l,$$

where $s_{ft}^l = L_{ft}/L_{it}$ is firm f 's labor-market share.

Interpretation. A bigger firm faces a steeper residual labor supply curve, so even without collusion larger firms have larger markdowns. This is the paper's **noncollusive lower bound**.

3.5.2 Full collusion

Under perfect collusion, the markdown becomes

$$m_{ft}^{\text{FC}} = 1 + \Psi^l.$$

Interpretation. Colluding firms internalize the full market labor supply, not just their own residual supply. The markdown no longer depends on individual firm size. This is the paper's **fully collusive upper bound**.

3.5.3 General conduct

With a conduct parameter $\tilde{\lambda}_{ft} \in [s_{ft}^l, 1]$,

$$m_{ft}^l = 1 + \tilde{\lambda}_{ft} \Psi^l.$$

Interpretation. The conduct parameter moves continuously between the Cournot benchmark and full collusion.

3.6 Cost-side identification of markdowns

From the materials first-order condition, the price markup is

$$m_{ft} = \frac{\beta_m}{a_{ft}^m},$$

where a_{ft}^m is the revenue share of materials.

Using the labor and materials conditions together gives the core identity:

$$m_{ft}^l = 1 + \tilde{\lambda}_{ft} \Psi^l = \frac{\beta_l a_{ft}^m}{\beta_m a_{ft}^l},$$

where a_{ft}^l is labor's revenue share.

Interpretation.

- The **left side** is the labor-supply-side markdown and depends on conduct.
- The **right side** is the **cost-side markdown** and does *not* require imposing a conduct assumption.

This is the paper's central trick: estimate the actual markdown from production and cost data, then compare it to the no-collusion and full-collusion bounds implied by labor supply.

3.7 Collusion index

The paper converts conduct into a 0-to-1 collusion index:

$$\hat{\lambda}_{ft} = \frac{m_{ft}^l - m_{ft}^{\text{NC}}}{m_{ft}^{\text{FC}} - m_{ft}^{\text{NC}}}.$$

- $\hat{\lambda}_{ft} = 0$: no collusion
- $\hat{\lambda}_{ft} = 1$: full collusion

Interpretation. This measures how far the actual markdown sits between the noncollusive and fully collusive bounds.

3.8 Estimation of production and labor supply

3.8.1 Production function

The production function is estimated by GMM using lagged inputs and **lagged agricultural wages** as an extra instrument:

$$E[u_{ft} \mid l_{f,t-1}, m_{f,t-1}, k_{ft}, w_{t-1}^{\text{agri}}] = 0.$$

Interpretation. Agricultural wages shift labor supply into coal mining but should not directly shift coal productivity. This helps identify production parameters beyond standard timing assumptions.

3.8.2 Labor supply

Labor supply is estimated at the market-year level using two labor-demand shifters:

- the **1871–75 coal demand shock** after the Franco-Prussian War,
- **cartel membership interacted with the postcartel period.**

Interpretation. These instruments move labor demand, not labor supply. That gives exogenous movement along the labor supply curve.

3.9 Counterfactual design

The paper studies two counterfactual environments:

1. **Exogenous coal prices.** Individual mines are tiny relative to the broader coal market. This gives a **lower bound** on the cartel's effect on wages and employment.
2. **Endogenous coal prices.** Coal demand is

$$P_{it} = Q_{it}^{\eta} y_{it},$$

with estimated inverse demand elasticity $\eta = -0.383$. This treats product and labor markets as aligned local markets and gives an **upper bound** on cartel effects.

Two benchmark worlds are compared with the actual cartel:

- **Cournot / no collusion;**
- **pre-1897 conduct frozen at its 1897 average.**

Interpretation. The first benchmark asks what would happen with no labor-market collusion. The second asks what the cartel added on top of the imperfectly competitive world that already existed before 1897.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- Empirical labor papers often estimate labor-market power while **imposing** a conduct model.
- Standard collusion models on the demand side must infer both **marginal cost** and **conduct** from output prices, which is difficult.
- Contemporary employer collusion is often illegal and unobserved, so direct measurement is hard.

This paper's move is to combine a production-cost approach with a labor supply model, so conduct is inferred from the gap between an observed markdown and theoretically implied conduct bounds.

4.2 What identifies the new approach

- The **production function** identifies output elasticities $(\beta_l, \beta_m, \beta_k)$.
- Those elasticities, together with input revenue shares, identify the **actual cost-side markdown**.
- The **labor supply elasticity** identifies the markdown that would arise under **no collusion** and under **full collusion**.
- Comparing the actual markdown to those bounds identifies the **degree of collusion**.

Key intuition. The method works because the paper gets the same economic object—the wage markdown—from two different sources:

1. a production-cost expression;
2. a labor-supply-and-conduct expression.

Their gap reveals conduct.

4.3 Instruments and exogenous variation

- **Production side instrument:** lagged Belgian agricultural wages.
- **Labor supply instruments:** the 1871–75 coal demand shock and cartel membership in the postcartel period.
- **Coal demand in the counterfactual:** estimated productivity is used as a cost shifter.

Interpretation.

- Agricultural wages shift labor supply into mining.
- Coal demand shocks and cartel restrictions shift labor demand.
- These sources of variation give separate leverage on production and labor supply.

4.4 Why the estimates are credible

- **Homogeneous employers within local labor markets** is empirically plausible because wages barely vary across firms within towns.
- **Adjustment-cost threat:** labor and material expenditure respond immediately to the 1871 demand shock, while capital adjusts with a lag; this supports the timing assumptions.
- **Factor-biased technical change threat:** weak evidence for labor-saving technology adoption in this setting; the timing of investment does not line up with the post-1897 markdown surge.
- **Unionization threat:** unions were weak, and stronger worker bargaining would tend to reduce the estimated employer collusion measure, not mechanically create it.
- **Cartel timing not hard-coded into the conduct test:** the paper argues the method could detect collusion even without observing the cartel itself.

4.5 Relation to the literature

- **Labor-market power:** adds collusion to a literature that often assumes monopsonistic competition or oligopsony without cooperation.
- **Conduct identification in IO:** combines demand-side conduct logic with production-side identification in the spirit of De Loecker–Warzynski and De Loecker–Scott.
- **Economic history:** quantifies employer collusion during industrialization rather than only documenting it qualitatively.

5 Tables and figures

5.1 Figure 1: output per worker, wages, and labor cost share

Read:

- Panel A shows that output per worker and daily wages move together until about the mid-1870s, then start to diverge.
- Panel B shows labor's cost share is fairly stable before the late 1890s, then falls after 1900.
- The key point is that raw wage-versus-output comparisons are **not enough** to diagnose monopsony or collusion, because capital accumulation and product-market conditions are moving too. The figure motivates the structural production model.

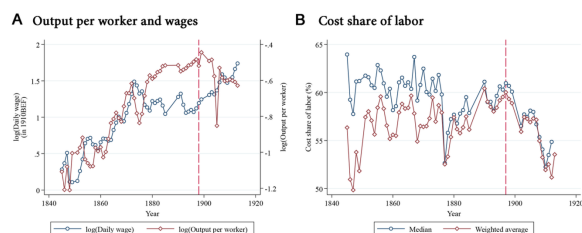


FIG. 1.—Output per worker, wages, and cost shares in Liège- and Namur-based coal mining, 1845–1913. *A*, The logarithm of total output divided by total days worked in Liège and Namur coal mines and the evolution of the logarithm of the average daily miner wage, weighted across mines by employment shares. *B*, The median and the average labor cost share of total expenditure, weighted by employment shares.

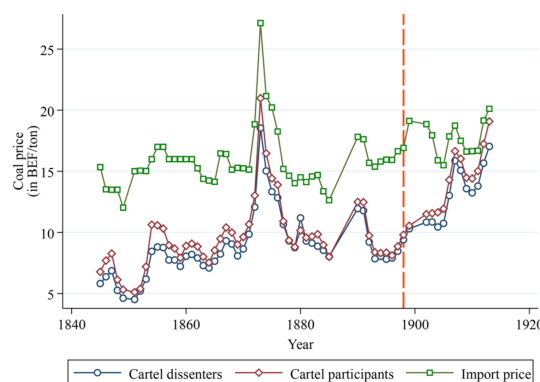


FIG. 2.—Prices and the Liège coal cartel, 1845–1913. The dashed vertical line represents the start of the coal cartel, the Syndicat de Charbonnages Liégeois.

5.2 Figure 2: prices and the 1897 cartel

Read:

- Before 1897, cartel participants price below the import price of coal.
- After the cartel begins, Liège coal prices rise toward the import-price ceiling.
- This is visual evidence that the cartel successfully restricted output and raised price.
- The same timing is suggestive for labor-market effects, because a downstream output cartel can reduce labor demand and increase markdowns.

5.3 Table 1: production function, markdowns, markups, labor supply

Read:

- Preferred GMM production estimates (free RTS) are roughly:

$$\begin{aligned}
 - \beta_l &= 0.699, \\
 - \beta_m &= 0.222,
 \end{aligned}$$

- $\beta_k = 0.153$,
- productivity persistence $\rho = 0.866$.
- This confirms coal mining is highly **labor intensive**.
- The estimated market-level inverse labor supply elasticity is **1.009**. That is economically large:
 - a monopsonist would pay about half the marginal revenue product of labor;
 - a firm with 10% market share would still have a meaningful markdown.
- Median wage markdown is **1.680**, implying wages about **40% below** marginal revenue product.
- Median coal price markup is only about **0.714**, so firms do not appear to have large downstream market power.
- Combining labor and product wedges still leaves firms with a positive total margin, so low price markups do not imply losses.

	log(Output)		log(Output)		log(Output)	
	Est.	SE	Est.	SE	Est.	SE
A. Production Function, log(Output)						
log(Labor) β^l	.794	.034	.699	.327	.661	.041
log(Materials) β^m	.275	.028	.222	.138	.237	.080
log(Capital) β^k	-.008	.140	.153	.075	.102	.088
Serial correlation TFP ρ			.866	.198	.853	.157
Method	OLS		GMM		GMM	
RTS	Free		Free		Fixed at 1.05	
R^2	.941		.938		.826	
Hansen J -test			2.34		2.72	
Hansen J -test p -value			.126		.255	
Number of firms	166		159		159	
Observations	4,480		4,005		4,005	
B. Markdowns and Markups						
Median markdown	1.541	.193	1.680	.450	1.486	.330
Average markdown	1.676	.224	1.828	.491	1.616	.361
Median markup	.884	.112	.714	.494	.763	.287
Average markup	.946	.120	.764	.535	.816	.315
Method	OLS		GMM		GMM	
RTS	Free		Free		Fixed at 1.05	
C. Labor Supply						
	log(Wage)		log(Wage)			
	Est.	SE	Est.	SE		
log(Employment)	.066	.006	1.009	.265		
Method	OLS		IV			
First-stage F -statistic			462			
Hansen J -test			5.92			
Hansen J -test p -value			.014			
Observations	1,990		1,990			
Firm-level elasticity	155.56		10.172			

NOTE.—Panels A and B are estimated at the firm-year level, and panel C is estimated at the market-year level. Standard errors (SEs) in panels A and B are block-bootstrapped with 200 iterations. Standard errors in panel C are estimated using the Driscoll and Kraay (1998) correction to allow for both cross-sectional (i.e., intratemporal) and intertemporal dependence, using the STATA command `ivreg2, draay(2)`.

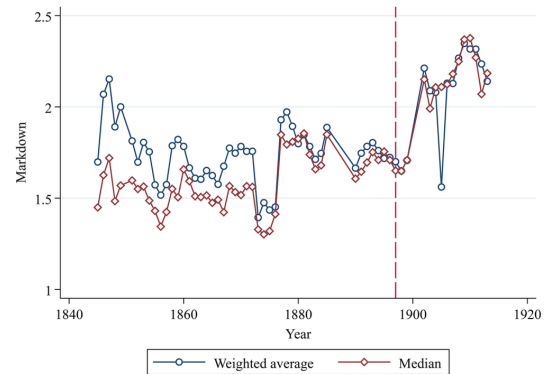


FIG. 3.—Evolution of average and median wage markdowns, 1845–1913. This graph shows the evolution of the weighted average (by employment) markdown and median wage markdown in Liège and Namur coal mines from 1845 to 1913.

5.4 Figure 3 and Table 2: evolution and correlates of markdowns

Read:

- Figure 3 shows a long-run pattern:
 - around **1.5** up to the 1870s,
 - around **1.7** in the 1880s–1890s,
 - above **2.2** after 1900.

- The post-1897 increase is persistent, not a short-lived shock.
- Table 2 shows wage markdowns are about **11.2% higher** at employers' association firms and **8.0% higher** at cartel firms.
- Pre-1898, employers' association membership is strongly associated with higher markdowns; after 1897 that association disappears.

Interpretation. Informal collusion via employers' associations seems important earlier, but after 1897 the formal cartel becomes the dominant collusive device.

	log(Markdown)		log(Markdown)	
	Est.	SE	Est.	SE
A. Markdown Correlations				
1(Employers' association)	.112	.052		
1(Cartel)	.080	.041		
1(1855 ≤ year < 1865)			-.021	.039
1(1865 ≤ year < 1875)			-.020	.038
1(1875 ≤ year < 1885)			.059	.045
1(1885 ≤ year < 1895)			.108	.047
1(1895 ≤ year < 1905)			.196	.045
1(1905 ≤ year < 1915)			.422	.054
Year fixed effects		Yes		No
R ²	.094		.076	
Observations	4,432		4,705	
B. Employers' Association, Pre- vs. Postcartel				
1(Employers' association)	.132	.042	-.058	.091
Period	1845-97		1898-1913	
R ²	.094		.130	
Observations	3,737		695	

NOTE.—Panels A and B are both estimated at the firm-year level. The reference category for the time dummies in panel A is the period between 1845–1859. Block-bootstrapped standard errors (SEs) are computed using 200 iterations.

	log(Markdown)					
	Est.	SE	Est.	SE	Est.	SE
A. All Firms						
log(Labor market share)	.044	.001	.055	.003	.051	.004
Fixed effects	None		Market		Market × year	
R ²	.067		.192		.550	
Observations	4,671		4,671		4,671	
B. Noncartel Firms						
log(Labor market share)	.037	.000	.053	.005	.065	.005
Fixed effects	None		Market		Market × year	
R ²	.046		.180		.561	
Observations	3,183		3,183		3,183	
C. Cartel Firms						
log(Labor market share)	.043	.001	.008	.002	-.004	.002
Fixed effects	None		Market		Market × year	
R ²	.063		.188		.793	
Observations	1,472		1,472		1,472	

NOTE.—We regress log markdowns on the log labor market employment share at the firm-year level for all firms (panel A), firms outside the cartel (panel B), and firms in the cartel (panel C). We control for a linear time trend and either no, market, or market × year fixed effects. The sample sizes of panels B and C add up to 4,655 because the cartel information is unobserved for 16 observations. Standard errors (SEs) are block-bootstrapped with 200 iterations.

5.5 Table 3: size-markdown correlations

Read:

- For **noncartel firms**, larger labor-market share predicts higher markdowns, even with market-year fixed effects.
- For **cartel firms**, that size-markdown relationship disappears once market controls are included.
- This is exactly what the model predicts:
 - without full collusion, larger firms exploit steeper residual labor supply;
 - with collusion, markdowns become much more equalized within the market.

5.6 Figure 4: collusive vs. noncollusive markdowns and the collusion test

Read:

- Panel A decomposes the actual markdown into where it lies relative to the no-collusion and full-collusion bounds.
- Before 1897, the actual markdown is above the noncollusive lower bound but below the fully collusive upper bound.

- After the cartel, the actual markdown moves up to the **fully collusive bound**.
- Panel B shows the estimated collusion index. Up to 1900, the paper cannot reject zero collusion; from **1901 onward** it generally can.

Interpretation. This is the central figure of the paper: it says the sharp post-1900 rise in markdowns is not just “more monopsony,” but specifically **more collusion**.

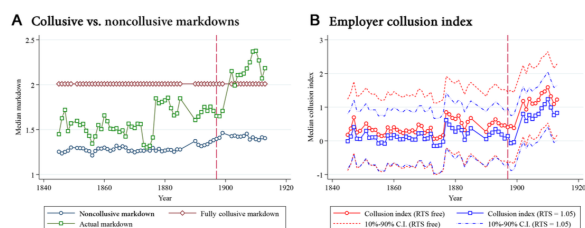


FIG. 4.—Employer collusion estimates, 1845–1913. A, The median markdown over time, along with the median of the lower and upper markdown bounds under no and full collusion. B, The median collusion index together with block-bootstrapped confidence intervals between 1845 and 1913. Two hundred bootstrap iterations are used.

TABLE 4
EFFECTS OF EMPLOYER COLLUSION

	Comparison: Cartel vs.	
	Cournot	Pre-1898 Conduct
A. Exogenous Price		
Relative wage change	−.103	−.059
Relative employment change	−.102	−.059
B. Endogenous Price		
Relative wage change	−.251	−.167
Relative employment change	−.249	−.166
Relative price change	.174	.100
Relative output change	−.283	−.195

NOTE.—Panel A summarizes the wage and employment effects of moving from the fully collusive coal cartel to either Cournot competition or to the estimated level of collusion prior of the cartel introduction, assuming exogenous coal prices. Panel B does the same for the model that allows for endogenous coal prices.

5.7 Table 4: counterfactual effects of the cartel

Read:

- **Exogenous coal price model:**
 - versus Cournot: wages −10.3%, employment −10.2%;
 - versus observed pre-1898 conduct: wages −5.9%, employment −5.9%.
- **Endogenous coal price model:**
 - versus Cournot: wages −25.1%, employment −24.9%, prices +17.4%, output −28.3%;
 - versus observed pre-1898 conduct: wages −16.7%, employment −16.6%, prices +10.0%, output −19.5%.
- The exogenous-price case is a lower bound and the endogenous-price case an upper bound.
- The key substantive result is that the cartel hurt workers a lot even under the conservative benchmark.

5.8 Sensitivity analysis

Read:

- **Adjustment frictions** are unlikely to explain the result because labor and materials react immediately to the 1871 demand shock, while capital reacts later.

- **Factor-biased technical change** is also an unlikely explanation because the timing of mechanization does not match the timing of the collusion surge, and machinery use is weakly correlated with markdowns.
- **Unionization and democratization** do not seem to drive the main pattern; worker organization was too weak in this period to overturn the employer-side explanation.

6 Short summary

- **Method.** Estimate wage markdowns from production and cost data, then compare them to model-based no-collusion and full-collusion bounds from labor supply.
- **Identification.** Production-function estimation gives actual markdowns; labor-supply estimation gives the conduct bounds; their comparison identifies collusion.
- **Application.** Belgian coal firms, 1845–1913, with legal cartels and detailed production data.
- **Main result.** The big increase in markdowns after 1900 was driven by employer collusion associated with the 1897 Liège cartel; the cartel reduced wages and employment by about **6%–17%** relative to precartel conduct and **10%–25%** relative to no-collusion benchmarks.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that employer collusion can be identified using **production, cost, and wage data**, without needing direct observation of collusive agreements.
- In the Belgian coal industry, rising wage markdowns after 1900 were not mainly the result of stronger ordinary oligopsony; they were the result of **collusion**.
- The 1897 cartel substantially reduced both wages and employment.

7.2 Economic intuition

- A firm with labor-market power already pays workers less than marginal revenue product.
- When firms **collude**, they internalize the effect of their labor demand on each other's wage bills, so they restrict labor demand more aggressively.
- If firms also collude downstream through an output cartel, that can amplify upstream monopsony power.
- So collusion on the product market can spill over into stronger labor exploitation even when product differentiation is limited.

7.3 Takeaway

This paper is important because it turns “employer collusion” from a historical allegation into something that can be **measured structurally**. The main lesson is simple: wage markdowns should not automatically be interpreted as noncooperative monopsony power. A meaningful part can come from **collusion**, and in this case that collusive component explains the sharp deterioration in workers’ outcomes after 1900.